SHREEYAN RIJAL

Computational Materials Physicist | Prospective PhD Researcher — Photovoltaic & Energy Materials

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Research Profile

Computational materials physicist specializing in first-principles (DFT) modeling of photovoltaic and energy-conversion materials. Work spans electronic-structure calculations, spin-polarized transport, and device-level simulation to study stability, efficiency, and optoelectronic behavior in perovskite solar cells, doped semiconductors, and van der Waals heterostructures. Author of four first-author peer-reviewed publications, with five further manuscripts submitted or under revision, and currently studying iodine-vacancy trap states and their passivation at $CsPbI_3$ surfaces. Seeking a fully funded PhD position to develop computational and data-driven methods for next-generation solar cell and quantum/energy materials.

Education

B.Sc. Physics **2021–2025**

St. Xavier's College, Tribhuvan University • Kathmandu, Nepal

Relevant Coursework: Quantum Mechanics, Nuclear & Solid State Physics, Mathematical Physics & Classical Mechanics, Numerical Methods, Computer Programming, Linear Algebra, Differential Equations, Statistics

Research Experience

- Investigating how an iodine vacancy at the $CsPbI_3$ (001) surface creates a trap state, and which small passivator molecules could remove it, using DFT in Quantum ESPRESSO with ORCA pre-screening of molecules; combined with a literature review for publication.
- Modeled perovskite solar-cell stability under simulated thermal and optical stress with device-level simulations in OghmaNano, identifying parameter regimes that improve stability and efficiency.
- Carried out spin-polarized first-principles calculations of Mn-doped silicon, resolving impurity-induced near-metallicity, and characterized its enhanced infrared optical activity, with direct relevance to spintronic device design.
- Modeled strain-engineered optoelectronic and thermoelectric response of two van der Waals heterostructures: Janus $MoSSe/Ti_2CO_2$ and Zr_2CO_2/blue phosphorus.
- Analyzed large geomagnetic and ionospheric observational datasets with time–frequency methods to characterize prompt penetration electric fields.
- Built Python-based scientific computing workflows for data analysis and machine-learning-assisted interpretation of simulation and observational datasets.
- Collaborated across multi-institutional research teams, translating results into four first-author peer-reviewed publications.

Publications (Peer-Reviewed, First Author)

1. **Rijal, S.** Strain-tunable band alignment, optical anisotropy, and thermoelectric response of a Janus $MoSSe/Ti_2CO_2$ van der Waals heterostructure. *Journal of Physics and Chemistry of Solids*, 114075. (Sole Author) DOI: 10.1016/j.jpcs.2026.114075
2. **Rijal, S.**, Chaudhary, P., Sharma, S., Thada Magar, M.T., Regmi, B., Adhikari, B., et al. Time-frequency characterization of prompt penetration electric fields across distinct geomagnetic regimes. *Advances in Space Research*. DOI: 10.1016/j.asr.2026.06.060
3. **Rijal, S.**, Jha, V.K., et al. Impurity-induced near-metallicity, spin-polarized transport, and enhanced infrared optical activity in Mn-doped silicon: A spin-polarized first-principles study. *Physica B: Condensed Matter*, 739, 418967. DOI: 10.1016/j.physb.2026.418967
4. **Rijal, S.**, Sah, N., Chaudhary, P., & Jha, V.K. Enhancing the stability and efficiency of perovskite solar cells by optimization of electrical parameter under thermal and optical stress. *Optical and Quantum Electronics* (Springer), 58, 288. DOI: 10.1007/s11082-026-08876-3

Manuscripts Submitted or Under Revision

1. **Rijal, S.**, et al. Relativistic Orbital Hybridization, Anisotropic Charge Transport and Dielectric Screening in Lead-Free Layered Double Perovskite $Cs_4MnSb_2Cl_{12}$ for Autonomous Indoor Photovoltaics. *Submitted.*
2. **Rijal, S.**, et al. Directional Spin–Orbit Coupling and Nonlinear Optical Response in a Lead-Free Bismuth–Silver Perovskite Cluster. *Submitted.*
3. **Rijal, S.**, et al. Multi-Instrument Evidence for Composition-Driven Nonlinear Ionosphere–Thermosphere Scaling Across G1–G5 Solar Cycle 25 Geomagnetic Storms. *Space Weather*; revision requested, in revision.
4. **Rijal, S.** Site-Specific Antioxidant Reactivity of Luteolin: A Multi-Descriptor DFT Study. *Submitted.*
5. **Rijal, S.**, et al. PM2.5 Divergence Across Asian Capitals: Atmospheric Drivers, Machine Learning Forecasting, and Governance Implications. *Submitted.*

In Preparation

1. **Rijal, S.** Iodine-Vacancy Trap States and Small-Molecule Passivation at the $CsPbI_3$ (001) Surface: A First-Principles Study and Literature Review. *Working title.*
2. **Rijal, S.** Strain-Engineered Optoelectronic and Thermoelectric Properties of Zr_2CO_2/Blue Phosphorus van der Waals Heterostructure: A First-Principles Study.

Technical Skills

DFT & Simulation: Quantum ESPRESSO, BoltzTraP2, Gaussian, ORCA, OghmaNano (device simulation)

Programming & Analysis: Python (scientific computing, machine learning, automation), MATLAB, C, LaTeX

Physics Areas: Condensed matter and semiconductor physics, electronic structure, spintronics, photovoltaics, optoelectronics, thermoelectrics

Leadership & Mentorship

- **Research Mentor**, St. Xavier's College (2023–Present) — Guide junior undergraduates into DFT research and scientific writing.
- **LSRA Mentor** (2024–Present) — Mentor a candidate for the Fathers Locke & Stiller Research Award.
- **Batch Representative**, Physics Department, St. Xavier's College (2023) — Represented the 2021 undergraduate cohort.
- **Library Council Representative**, Physics Department, St. Xavier's College (2021–2025) — Sustained institutional leadership and service throughout undergraduate study.

Languages

Nepali (Native) • English (Fluent, Professional Proficiency) • Hindi (Fluent, Professional Proficiency)

References

Dr. Vinaya Kumar Jha — Research Supervisor
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Dr. Binod Adhikari — Research Supervisor; Former Head of Department
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Dr. Basu Dev Ghimire — Head of Department; Instructor
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